

AliBahmani IronOre Process Simulator

Industrial Mineral Processing Simulation and Engineering Software

Software Version	1.0.0
Model Versions	MagneticModel v1, ScreenModel v1, CrusherModel v1, RecycleSolver v1, MassB
Project Name	Dry Iron Ore Beneficiation Plant - 600 t/h
Simulation ID	2732A98C
Simulation Date	2026-09-18
Developer / Manufacturer	Ali Bahmani (■■■■ ■■■■■■)
Contact	09915420558
Simulation Status	CONVERGED

MODEL RESULT — this report presents model results, not site-verified results; site calibration data is required before trusting recovery predictions (Section 113).

1. Final Report

Metric	Value	Unit
Feed Rate	600.000	t/h
Feed Fe Grade	18.5	%
Final Product Rate	6.577	t/h
Final Product Grade	60.077	%
Final Tail Rate	593.421	t/h
Final Tail Grade	18.039	%
Mass Recovery	1.096	%
Fe Recovery	3.560	%
Fe Input	111.000	t/h
Fe Product	3.951	t/h
Fe Tail	107.048	t/h
Circulating Load	96.648	t/h
Circulating Load	16.108	%
Screen Load	149.268	t/h
Crusher Load	96.648	t/h
Daily Production	131.543	t/day
Monthly Production	3,420.108	t/month
Annual Production	43,409.064	t/year
Mass Balance Error	0.000	%
Fe Balance Error	0.001	%
Convergence Status	CONVERGED	

2. Stream-by-Stream Results

Stream ID	Dry t/h	Wet t/h	Fe %	Fe t/h	Status
S-GYR	600.00	603.02	18.500	111.00	OK
S-PRE1-IN	100.00	100.50	18.500	18.50	OK
S-PRE1-MAG	8.77	8.81	41.654	3.65	OK
S-PRE1-TAIL	91.23	91.69	16.274	14.85	OK
S-PRE2-IN	100.00	100.50	18.500	18.50	OK
S-PRE2-MAG	8.77	8.81	41.654	3.65	OK
S-PRE2-TAIL	91.23	91.69	16.274	14.85	OK
S-PRE3-IN	100.00	100.50	18.500	18.50	OK
S-PRE3-MAG	8.77	8.81	41.654	3.65	OK
S-PRE3-TAIL	91.23	91.69	16.274	14.85	OK
S-PRE4-IN	100.00	100.50	18.500	18.50	OK

S-PRE4-MAG	8.77	8.81	41.654	3.65	OK
S-PRE4-TAIL	91.23	91.69	16.274	14.85	OK
S-PRE5-IN	100.00	100.50	18.500	18.50	OK
S-PRE5-MAG	8.77	8.81	41.654	3.65	OK
S-PRE5-TAIL	91.23	91.69	16.274	14.85	OK
S-PRE6-IN	100.00	100.50	18.500	18.50	OK
S-PRE6-MAG	8.77	8.81	41.654	3.65	OK
S-PRE6-TAIL	91.23	91.69	16.274	14.85	OK
S-MERGE	52.62	52.89	41.654	21.92	OK
S-RECYCLE-IN	96.65	97.13	39.893	38.55	OK
S-SCR-FEED	149.27	150.02	40.514	60.47	OK
S-SCR1-IN	49.76	50.01	40.514	20.16	OK
S-SCR1-UND	17.54	17.63	41.654	7.31	OK
S-SCR1-OV	32.22	32.38	39.893	12.85	OK
S-SCR2-IN	49.76	50.01	40.514	20.16	OK
S-SCR2-UND	17.54	17.63	41.654	7.31	OK
S-SCR2-OV	32.22	32.38	39.893	12.85	OK
S-SCR3-IN	49.76	50.01	40.514	20.16	OK
S-SCR3-UND	17.54	17.63	41.654	7.31	OK
S-SCR3-OV	32.22	32.38	39.893	12.85	OK
S-FINES	52.62	52.88	41.654	21.92	OK
S-OV-COARSE	96.65	97.13	39.893	38.56	OK
S-COARSE-OUT	96.65	97.13	39.893	38.56	OK
S-FINE-OUT	96.65	97.13	39.893	38.56	OK
S-FIN1-IN	26.31	26.44	41.654	10.96	OK
S-PRODUCT-1	3.29	3.31	60.077	1.98	OK
S-TAIL-1	23.02	23.14	39.022	8.98	OK
S-FIN2-IN	26.31	26.44	41.654	10.96	OK
S-PRODUCT-2	3.29	3.31	60.077	1.98	OK
S-TAIL-2	23.02	23.14	39.022	8.98	OK
S-TAIL-FIN	46.04	46.27	39.022	17.97	OK
S-TAIL-PRE	547.38	550.13	16.274	89.08	OK
S-TAIL	593.42	596.40	18.039	107.05	OK
S-PRODUCT	6.58	6.61	60.077	3.95	OK
S-FEED	600.00	603.02	18.500	111.00	OK

3. Warnings & Assumptions

- Separator capacity = CALIBRATION REQUIRED (no manufacturer/test data)

- GYRATORY capacity = CALIBRATION REQUIRED (no manufacturer data)
Assumptions: separator capacity/power/pole geometry are CALIBRATION REQUIRED or UNKNOWN; screen and crusher models use Tier-5 engineering correlations pending testwork calibration; month=30 d, year=365 d for period conversions.

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